

CHAPTER 2

Introduction to the Human Genome

Understanding the organization, variation, and transmission of the **human genome** is central to appreciating the role of genetics in medicine, as well as the emerging principles of genomic and personalized medicine. With the availability of the sequence of the human genome and a growing awareness of the role of genome variation in disease, it is now possible to begin to exploit the impact of that variation on human health on a broad scale. The comparison of individual genomes underscores the first major take-home lesson of this book—*every individual has his or her own unique constitution of gene products, produced in response to the combined inputs of the genome sequence and one's particular set of environmental exposures and experiences*. As pointed out in the previous chapter, this realization reflects what Garrod termed *chemical individuality* over a century ago and provides a conceptual foundation for the practice of genomic and personalized medicine.

Advances in genome technology and the resulting explosion in knowledge and information stemming from the **Human Genome Project** are thus playing an increasingly transformational role in integrating and applying concepts and discoveries in genetics to the practice of medicine.

CHROMOSOME AND GENOME ANALYSIS IN CLINICAL MEDICINE

Chromosome and genome analysis has become an important diagnostic procedure in clinical medicine. As described more fully in subsequent chapters, these applications include the following:

- ***Clinical diagnosis.*** Numerous medical conditions, including some that are common, are associated with changes in chromosome number or structure and require chromosome or genome analysis for diagnosis and genetic counseling (see Chapters 5 and 6).
- ***Gene identification.*** A major goal of medical genetics and genomics today is the identification of specific genes and elucidating their roles in health and disease. This topic is referred to repeatedly but is discussed in detail in Chapter 10.
- ***Cancer genomics.*** Genomic and chromosomal changes in somatic cells are involved in the initiation and progression of many types of cancer (see Chapter 15).
- ***Disease treatment.*** Evaluation of the integrity, composition, and differentiation state of the genome is critical for the development of patient-specific pluripotent stem cells for therapeutic use (see Chapter 13).
- ***Prenatal diagnosis.*** Chromosome and genome analysis is an essential procedure in prenatal diagnosis (see Chapter 17).

THE HUMAN GENOME AND THE CHROMOSOMAL BASIS OF HEREDITY

Appreciation of the importance of genetics to medicine requires an understanding of the nature of the hereditary material, how it is packaged into the human genome, and how it is transmitted from cell to cell during cell division and from generation to generation during reproduction. The human genome consists of large amounts of the chemical deoxyribonucleic acid (**DNA**) that contains within its structure the genetic information needed to specify all aspects of embryogenesis, development, growth, metabolism, and reproduction—essentially all aspects of what makes a human being a functional organism. Every nucleated cell in the body carries its own copy of the human genome, which contains, depending on how one defines the term, approximately 20,000 to 50,000 **genes** (see Box later). Genes, which at this point we consider simply and most broadly as functional units of genetic information, are encoded in the DNA of the genome, organized into a number of rod-shaped organelles called **chromosomes** in the nucleus of each cell. The influence of genes and genetics on states of health and disease is profound, and its roots are found in the information encoded in the DNA that makes up the human genome.

Each species has a characteristic chromosome complement (**karyotype**) in terms of the number, morphology, and content of the chromosomes that make up its genome. The genes are in linear order along the chromosomes, each gene having a precise position or **locus**. A **gene map** is the map of the genomic location of the genes and is characteristic of each species and the individuals within a species.